~~Experiment~~ Assignment-4

Problem Statement:

Write a C program to store first year percentage of students in an array. Write a function for sorting array of floating-point numbers in ascending order using Bucket Sort and display top five scores.

Objective:

1) Understand Bucket Sorte
2) Implement Bucket Sort using functions
3) Sort floating-point percentages
4) Display top five scores
5) Analyze complexity.

Implementation:

- Platform
  - 64-bit Open source Linux or its derivative
  - Open Source C Programing tool like gcc / Eclipse Editor.

- Test conditions

Test Case	Input	Expected O/P
TC1	One percentage	Stored succeffully
TC2	Multiple percentages	Stored correctly
TC3	Random floting values	Ascendng Ordr
TC4	Duplicated values	Duplicates retained
TC5	< 5 students	Display avaible scors
TC6	>= 5 students	Display top five scores

PSeudo Code

Algorithm BucketSort(arr, n):
1. Create n empty buckets (linked lists or dynamic arrays)
2. For i = 0 to n-1:
   a. Calculate bucket index: index = floor(n * arr[i])
   b. Insert arr[i] into bucket[index]
3. For i = 0 to n-1:
   a. Sort individual bucket[i] using Insertion Sort (or any standard sorting)
4. Concatenate/Merge all buckets sequentially into the original array arr.

Algorithm Main:
1. Read the number of students, n.
2. Read n percentage values
3. Call BucketSort(arr, n)
4. Print the sorted array in ascending order
5. Display top five scores:
   a. If n < 5:
      Print all n scores starting from index (n-1) down to 0.
   b. Else:
      Print elements from index (n-1) down to (n-5)

Time Complexity
- Best Case O(n+k)
- Average Case: O(n+k)
- Worst Case: $O(n^2)$
- Space Complexity O(n+k)

Conclusion:
Thus, implemented different searching and sorting methods on the student database. This system is able to perform sorting under different cases.

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FAQs

1. ~~What is the meaning of da~~

1) What is Bucket Sort.

Ans Bucket Sort is a distribution-based comparison sort that works by partitioning an array into a finite number of buckets. Each bucket is then sorted individually - either using a different sorting algorithm or recursively applying bucket sort - before the buckets are concatenated

2) Why use it for floating point data?
Normalized float values directly map to bucket indices, enabling linear average time $O(n)$ for uniformly distributed data.

3) What is a bucket?
A temporary container (like a linked list or dynamic array) used to store numbers falling within a specific range.

4) What is the worst case complexity?
$O(n^2)$ which occurs when all elements end up in a single bucket.

5) What are its applications?
Sorting uniformly distributed continuous numbers (like test scores), histogram generation, and parallel processing.

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